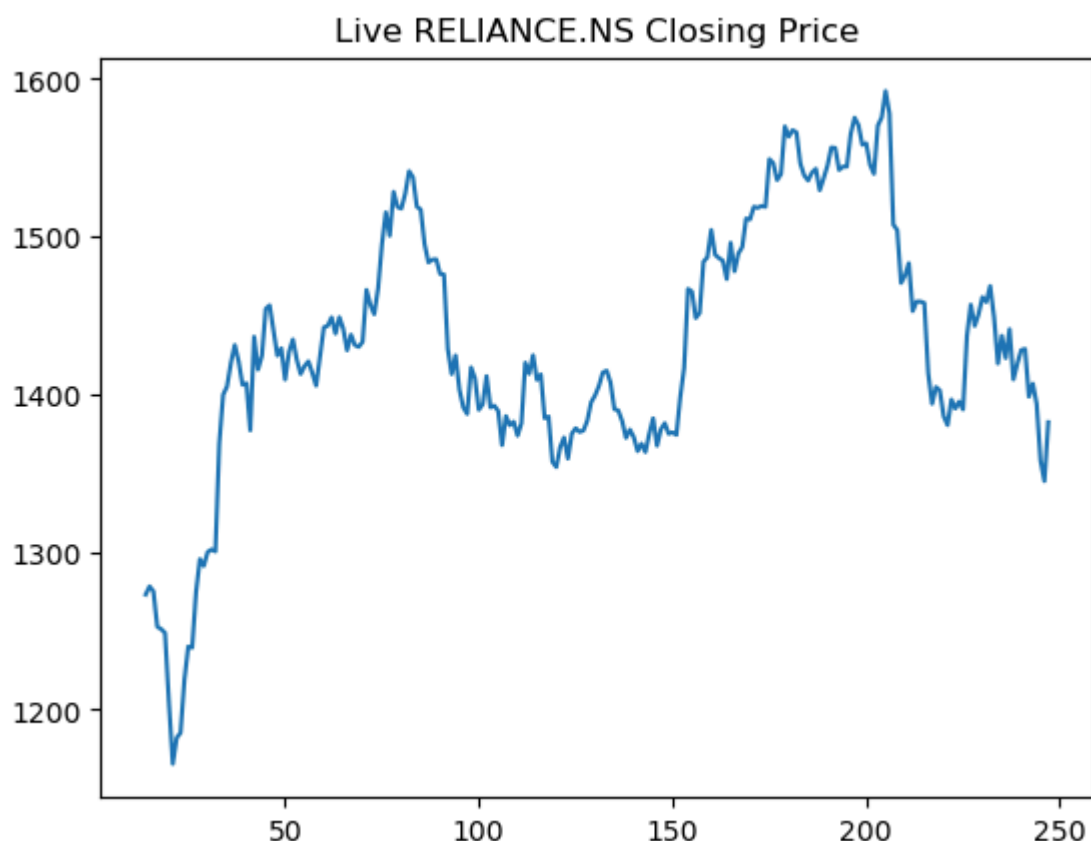


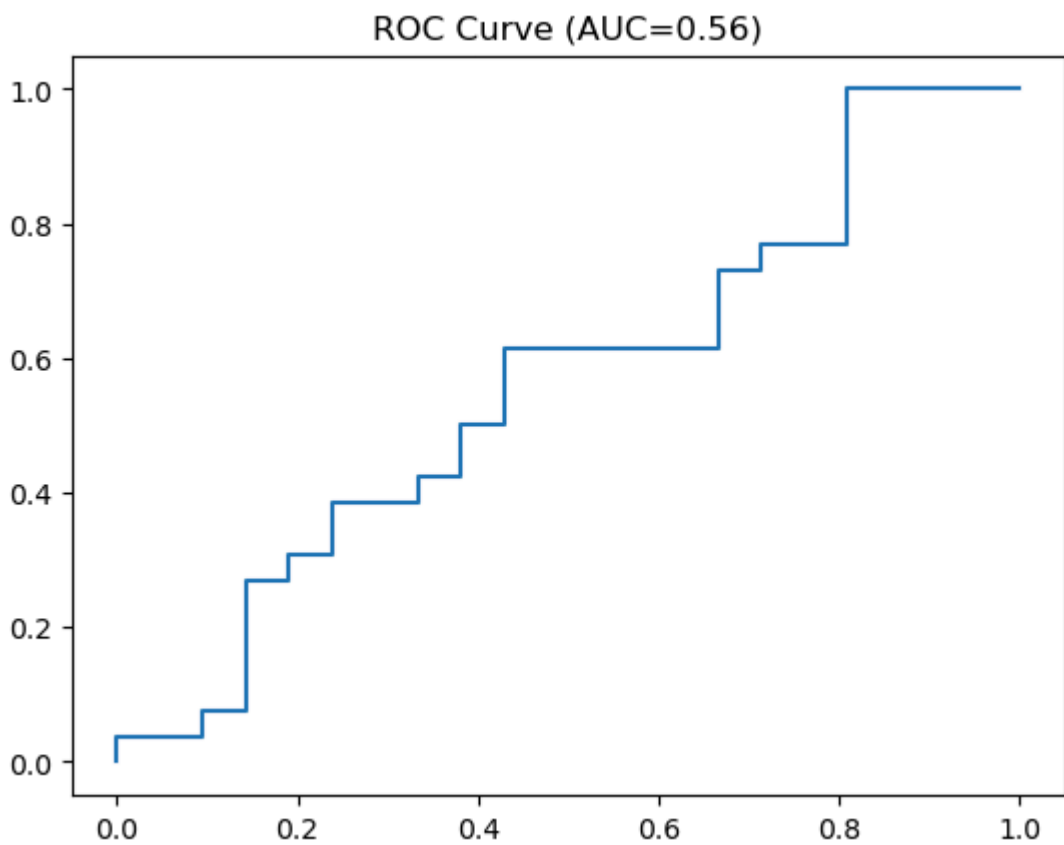
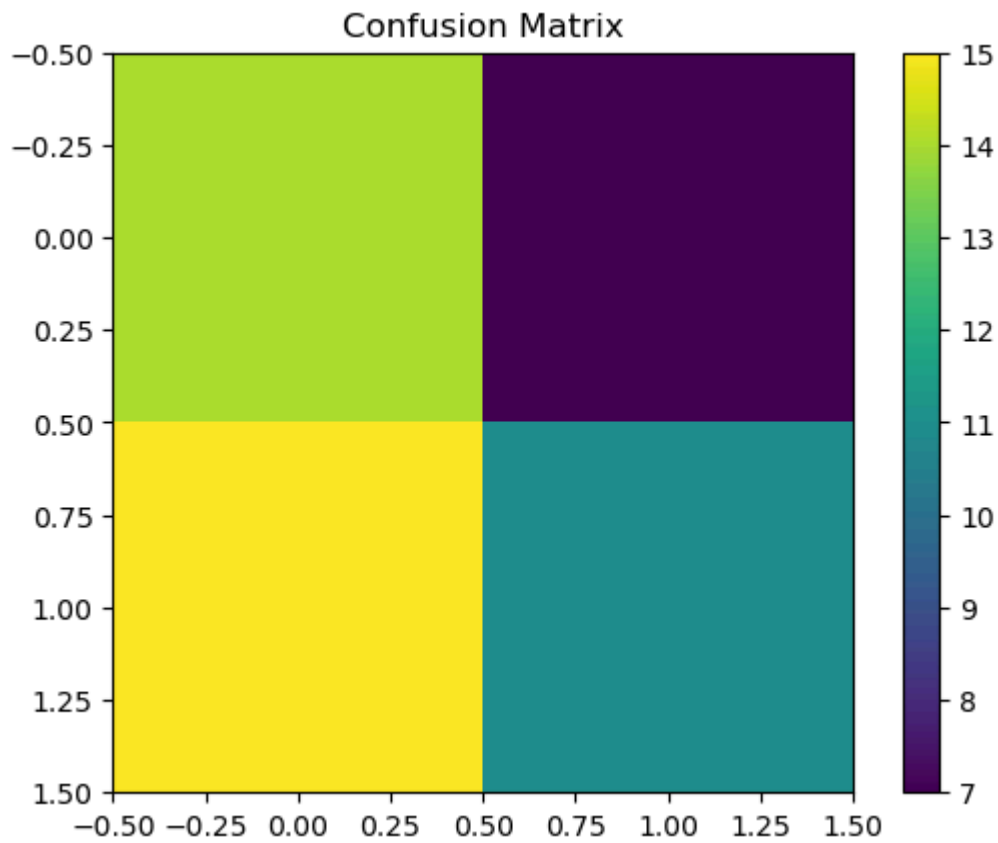
```

In [2]: import pandas as pd, numpy as np, requests, matplotlib.pyplot as plt
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import confusion_matrix, roc_curve, auc
symbol="RELIANCE.NS"
url=f"https://query1.finance.yahoo.com/v8/finance/chart/{symbol}?range=1y&interval=1d"
r=requests.get(url,headers={"User-Agent":"Mozilla/5.0"}).json()
q=r["chart"]["result"][0]["indicators"]["quote"][0]
data=pd.DataFrame(q)[["open","high","low","close","volume"]]
data.columns=["Open","High","Low","Close","Volume"]
data=data.dropna()
data["MA5"]=data["Close"].rolling(5).mean()
data["MA10"]=data["Close"].rolling(10).mean()
d=data["Close"].diff()
data["RSI"]=100-100/(1+d.clip(lower=0).rolling(14).mean()/(
d.clip(upper=0).rolling(14).mean()))
data["Target"]=(data["Close"].shift(-1)>data["Close"]).astype(int)
data.dropna(inplace=True)
X=StandardScaler().fit_transform(data[["Open","High","Low","Close","Volume","MA5"]])
y=data["Target"]
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
model=LogisticRegression(max_iter=3000).fit(X_train,y_train)
pred=model.predict(X_test)
prob=model.predict_proba(X_test)[:,1]
print("Live Accuracy:",model.score(X_test,y_test))
plt.figure(); plt.plot(data["Close"]); plt.title("Live RELIANCE.NS Closing Price")
plt.figure(); plt.imshow(confusion_matrix(y_test,pred)); plt.title("Confusion Matrix")
plt.colorbar(); plt.show()
fpr,tpr,_=roc_curve(y_test,prob)
plt.figure(); plt.plot(fpr,tpr); plt.title("ROC Curve (AUC=%.2f)"%auc(fpr,tpr));

```

Live Accuracy: 0.5319148936170213





In []: